WaveFlow-UIE: A Single-Pipeline Frequency-Guided Flow Model for Underwater Image Restoration

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April 2026

Abstract

Underwater images often suffer from color distortion, haze, low contrast, and loss of fine details, which can reduce both visual quality and the reliability of downstream vision tasks. We propose WaveFlow-UIE, a single-pipeline wavelet-domain model for underwater image restoration. Unlike WF-Diff, which uses a coarse enhancement network followed by a separate diffusion-based refinement module, WaveFlow-UIE directly learns a flow from degraded wavelet representations to clean wavelet representations. The model decomposes an input image using the discrete wavelet transform, estimates physics-inspired degradation cues such as transmission and ambient light, and uses these cues to condition a velocity network trained with flow matching. Frequency-aware, perceptual, color, and physics consistency losses are used to improve both low-frequency color correction and high-frequency detail preservation. Experiments on multiple underwater benchmarks show that WaveFlow-UIE provides competitive restoration quality while simplifying the restoration pipeline and reducing the dependence on a separate diffusion refinement stage.

1 Introduction

Underwater image restoration is an important problem in computer vision because images captured in underwater environments often suffer from severe visual degradation. Light absorption and scattering in water can cause color distortion, low contrast, haze, blur, and loss of fine texture details. These degradations make underwater images harder to interpret and can reduce the performance of downstream vision systems such as underwater object detection, tracking, robotic navigation, marine inspection, and scientific exploration.

Recent deep learning methods have improved underwater image enhancement by learning a mapping from degraded images to visually clearer images. Among these methods, WF-Diff (Zhao et al., 2024) is a strong frequency-domain restoration framework. Its key idea is that underwater degradation can be better handled by separating an image into frequency components. Using the discrete wavelet transform, low-frequency components mainly represent color, illumination, and global structure, while high-frequency components mainly represent edges, textures, and fine details. WF-Diff uses this observation to perform wavelet-based Fourier information interaction, followed by a diffusion-based frequency residual adjustment module.

Although WF-Diff achieves strong restoration quality, its architecture follows a two-stage design. First, WFI2-net generates a coarse enhanced output in wavelet space. Then, FRDAM applies diffusion-based refinement to adjust the remaining low- and high-frequency residual errors. While effective, this separation introduces additional architectural complexity and increases inference cost due to the diffusion refinement stage. This motivates the question of whether underwater image restoration can be reformulated as a more direct single-pipeline process.

In this work, we propose WaveFlow-UIE, a single-pipeline wavelet-domain model for underwater image restoration. Instead of first producing a coarse output and then applying a separate diffusion refinement module, WaveFlow-UIE directly learns a transformation from degraded wavelet representations toward clean wavelet representations. The model uses a conditional velocity network to predict the restoration direction in wavelet space. To better guide the enhancement process, we also introduce a lightweight physics prior branch that estimates underwater image formation cues, including transmission and ambient light. These cues provide additional conditioning that helps the model reason about underwater color degradation and haze.

Our method keeps the main insight of WF-Diff, namely that frequency information is important for underwater restoration, but simplifies the restoration process into a unified flow-based pipeline. During training, the model is supervised using a combination of flow matching loss, frequency-weighted loss, perceptual loss, color consistency loss, and physics consistency loss. This design encourages the model to improve both low-frequency color correction and high-frequency detail preservation.

The main contributions of this work are summarized as follows:

- We reformulate WF-Diff-style underwater image restoration as a single-pipeline wavelet-domain flow matching problem.
- We introduce physics prior conditioning to guide the restoration process using underwater image formation cues.
- We apply frequency-aware training losses to balance low-frequency color restoration and high-frequency detail enhancement.
- We provide a framework for comparing the proposed WaveFlow-UIE model with the original WF-Diff baseline under consistent evaluation settings.

2 Related Work

2.1 Underwater image restoration

Underwater image restoration aims to recover visually clear images from degraded underwater observations. Images captured underwater often suffer from color cast, low contrast, haze, and loss of fine details due to light absorption and scattering. Early methods commonly relied on physical image formation models and hand-crafted priors to estimate degradation factors such as transmission, background light, or color attenuation. While these methods are interpretable, they often struggle in complex real-world underwater scenes where lighting conditions, water types, and scene depths vary significantly.

Recent learning-based methods have improved restoration quality by learning mappings from degraded underwater images to clean reference images. Convolutional neural networks, attention-based models, and GAN-based methods have been used to enhance color, contrast, and texture details. Representative underwater enhancement methods include Water-Net, UWCNN, Ucolor, U-shape, and DM-water. However, many of these methods operate mainly in the spatial image domain, which can limit their ability to explicitly model frequency-specific degradation such as color shifts and texture loss.

2.2 Frequency-domain image restoration

Frequency-domain representations have become useful in low-level vision because they allow an image to be decomposed into components with different visual meanings. In underwater image restoration, low-frequency components are closely related to color, illumination, and global structure, while high-frequency components are related to edges, textures, and fine details. This separation is useful because underwater degradation affects these components differently.

WF-Diff (Zhao et al., 2024) showed that wavelet and Fourier information can be effectively used for underwater image restoration. By applying discrete wavelet transform, WF-Diff separates an image into low-frequency and high-frequency sub-bands. It then uses Fourier information interaction to enhance frequency components before reconstructing the image. This motivates our work, since we also preserve the idea that frequency-domain processing is important for underwater restoration. However, instead of using the same two-stage enhancement and diffusion refinement design, we reformulate the restoration process as a single wavelet-domain flow pipeline.

2.3 Diffusion and flow-based restoration

Diffusion models have achieved strong results in image generation and restoration by learning to reverse a noise corruption process. In image restoration, diffusion models can be used to refine degraded images and recover visually realistic details. WF-Diff applies this idea through its Frequency Residual Diffusion Adjustment Module, which learns residual corrections for low- and high-frequency components after an initial coarse restoration stage.

Although diffusion-based refinement can improve visual quality, it often requires multiple sampling steps, which increases inference cost. Flow matching and rectified flow methods provide an alternative formulation by learning a direct transformation path between two distributions. Instead of repeatedly denoising from a highly noisy state, a flow-based model learns a velocity field that moves samples from a source representation toward a target representation. In our work, we use this idea in wavelet space: the model learns how to move degraded underwater wavelet features toward clean wavelet features. This allows us to replace the separate coarse-enhancement and diffusion-refinement stages with a unified single-pipeline restoration model.

3 Proposed Method

3.1 Overview

We propose WaveFlow-UIE, a single-pipeline wavelet-domain model for underwater image restoration. The main goal is to restore a degraded underwater image by directly learning a transformation from its degraded frequency representation to the corresponding clean frequency representation. Unlike WF-Diff, which first produces a coarse enhanced output and then applies a separate diffusion-based residual refinement module, our method performs restoration through one unified flow-based pipeline.

Given a degraded underwater image, WaveFlow-UIE first applies a discrete wavelet transform to decompose the image into low-frequency and high-frequency components. A physics prior branch then estimates underwater degradation cues, including transmission and ambient light. These cues are used as conditioning information for a velocity network, which predicts how the degraded wavelet representation should move toward the clean wavelet representation. Finally, the restored wavelet representation is converted back to the image domain using the inverse discrete wavelet transform.

3.2 Wavelet-domain representation

Underwater degradation affects different visual components in different ways. Color distortion and illumination changes are mainly represented in low-frequency components, while edges, textures, and fine details are mainly represented in high-frequency components. Therefore, we perform restoration in wavelet space instead of directly operating only in RGB image space.

Given a degraded underwater image $x \in \mathbb{R}^{3\times H\times W}$, we apply the Haar discrete wavelet transform:

$$w_x = \mathrm{DWT}(x), \tag{1}$$

where $w_x \in \mathbb{R}^{12\times \frac{H}{2}\times \frac{W}{2}}$. The 12 channels correspond to four wavelet sub-bands for each RGB channel: one low-frequency sub-band and three high-frequency sub-bands. Similarly, for a clean ground-truth image y, we compute:

$$w_y = \mathrm{DWT}(y). \tag{2}$$

This representation allows the model to learn restoration in a frequency-aware manner while still preserving all information needed to reconstruct the final image through the inverse wavelet transform.

3.3 Physics prior conditioning

Underwater image degradation is strongly related to the physical properties of light propagation in water. Inspired by underwater image formation models, we include a lightweight physics prior branch that estimates degradation cues from the input image. Specifically, the branch predicts a transmission map t and an ambient light map A, which are related to the simplified underwater formation model:

$$x \approx y \cdot t + A \cdot (1 - t). \tag{3}$$

The transmission map describes how much scene information remains visible after light travels through water, while the ambient light map represents the color cast and veiling light caused by the underwater medium. These estimated maps are downsampled to the wavelet resolution and concatenated into a physics conditioning tensor:

$$p = [t, A]. \tag{4}$$

This conditioning provides the restoration network with additional information about the type of underwater degradation present in the image.

3.4 Wavelet-domain flow matching

Instead of learning a separate coarse enhancement stage followed by diffusion residual refinement, WaveFlow-UIE learns a direct flow from degraded wavelet features to clean wavelet features. During training, we sample a time value $\tau \sim \mathcal{U}(0, 1)$ and construct an interpolated wavelet representation:

$$w_\tau = (1 - \tau)w_x + \tau w_y. \tag{5}$$

The target velocity is the direction from the degraded wavelet representation to the clean wavelet representation:

$$v^* = w_y - w_x. \tag{6}$$

A conditional velocity network v_θ is trained to predict this direction:

$$\hat{v} = v_\theta(w_\tau, \tau, w_x, p), \tag{7}$$

where w_τ is the current wavelet state, τ is the flow time, w_x is the degraded wavelet condition, and p is the physics prior condition.

The predicted clean wavelet representation is obtained as:

$$\hat{w}_y = w_x + \hat{v}. \tag{8}$$

The restored RGB image is then reconstructed using the inverse discrete wavelet transform:

$$\hat{y} = \mathrm{IDWT}(\hat{w}_y). \tag{9}$$

3.5 Training objective

The model is trained using a combination of losses that encourage accurate flow prediction, frequency-aware restoration, perceptual quality, color consistency, and physical consistency.

First, we use a conditional flow matching loss to supervise the predicted velocity:

$$\mathcal{L}_{\mathrm{cfm}} = \|\hat{v} - v^*\|_2^2. \tag{10}$$

Second, we use a frequency-weighted loss in wavelet space. Since high-frequency components are important for preserving edges and textures, we assign a larger weight to high-frequency errors:

$$\mathcal{L}_{\mathrm{freq}} = \|\hat{v}_{LL} - v^*_{LL}\|_1 + \alpha\|\hat{v}_{HF} - v^*_{HF}\|_1, \tag{11}$$

where LL denotes the low-frequency component, HF denotes the high-frequency components, and α controls the importance of high-frequency restoration.

We also include a perceptual loss to improve visual quality:

$$\mathcal{L}_{\mathrm{lpips}} = \mathrm{LPIPS}(\hat{y}, y), \tag{12}$$

and a Lab color loss to encourage color consistency:

$$\mathcal{L}_{\mathrm{lab}} = \|\mathrm{Lab}_{ab}(\hat{y}) - \mathrm{Lab}_{ab}(y)\|_1. \tag{13}$$

Finally, the physics prior branch is supervised using a consistency loss based on the underwater image formation model:

$$\mathcal{L}_{\mathrm{phys}} = \|x - (y \cdot t + A \cdot (1 - t))\|_2^2. \tag{14}$$

The total training objective is:

$$\mathcal{L}_{\mathrm{total}} = \lambda_{\mathrm{cfm}}\mathcal{L}_{\mathrm{cfm}} + \lambda_{\mathrm{freq}}\mathcal{L}_{\mathrm{freq}} + \lambda_{\mathrm{lpips}}\mathcal{L}_{\mathrm{lpips}} + \lambda_{\mathrm{lab}}\mathcal{L}_{\mathrm{lab}} + \lambda_{\mathrm{phys}}\mathcal{L}_{\mathrm{phys}}. \tag{15}$$

3.6 Inference

During inference, only the degraded underwater image is available. We first compute its wavelet representation w_x and physics prior condition p. The model then starts from the degraded wavelet representation and gradually updates it using the predicted velocity field:

$$w_{k+1} = w_k + \Delta t \cdot v_\theta(w_k, t_k, w_x, p), \tag{16}$$

where k is the inference step and Δt is the step size. After a fixed number of steps, the final wavelet representation is converted back to RGB space:

$$\hat{y} = \mathrm{IDWT}(w_K). \tag{17}$$

This inference procedure produces the restored underwater image through a single unified wavelet-domain flow pipeline, without requiring a separate coarse enhancement network and diffusion refinement module.

4 Experiments

4.1 Setup

Implementation details. Our network is implemented in PyTorch and trained on an NVIDIA GeForce RTX 5080 GPU; all evaluation runs are performed on an NVIDIA GeForce RTX 5070 Ti. We use the AdamW optimizer with $\beta_1 = 0.9$, $\beta_2 = 0.999$, and a weight decay of 1×10^{-4}. The patch size is 256×256 and the per-step batch size is 4 with 2 gradient accumulation steps, giving an effective batch size of 8; training is performed in mixed precision (FP16) with gradient clipping at 1.0. The initial learning rate is set to 2×10^{-4}, with a linear warm-up over the first 5,000 iterations followed by a cosine schedule decaying to $\eta_{\min} = 1 \times 10^{-6}$. We train three model variants under the same hyperparameters but on different sources: WaveFlow-UIEB on UIEB (Li et al., 2020; 200,000 iterations), WaveFlow-LSUI on LSUI (Peng et al., 2023; 2,000,000 iterations), and WaveFlow-BOTH on the union of UIEB and LSUI (2,000,000 iterations). The total training objective is a weighted combination of five terms with $\lambda_{\mathrm{cfm}} = 1.0$, $\lambda_{\mathrm{freq}} = 0.5$, $\lambda_{\mathrm{lpips}} = 0.1$, $\lambda_{\mathrm{lab}} = 0.05$, and $\lambda_{\mathrm{phys}} = 0.1$; within $\mathcal{L}_{\mathrm{freq}}$, high-frequency wavelet errors receive twice the weight of the low-frequency error ($\alpha = 2.0$). At inference, the wavelet state is updated for $K = 5$ Euler flow steps with step size $\Delta t = 1/K = 0.2$.

Datasets. We use the real-world UIEB dataset (Li et al., 2020) and the LSUI dataset (Peng et al., 2023) for training and evaluation. UIEB provides 890 underwater images with corresponding labels; we follow the standard split used by prior work, with 800 images for training and 90 for testing. LSUI is randomly partitioned into 3,879 training images and 400 testing images. In addition, to verify the cross-dataset generalization of WaveFlow-UIE, we evaluate on UFO-120 (Islam et al., 2020a) and EUVP (Islam et al., 2020b), and on the non-reference benchmarks U45 (Li et al., 2019) and the challenging-set C60.

Comparison methods. We conduct a comparative analysis between WaveFlow-UIE and six state-of-the-art underwater image enhancement methods: WF-Diff (Zhao et al., 2024), SCNet (Fu et al., 2022), UIE-WD (Ma et al., 2022), UIEC^2-Net (Wang et al., 2021), Water-Net (Li et al., 2020), and U-shape (Peng et al., 2023). To ensure a fair and rigorous comparison, we use the official released checkpoints and inference code for each method, and adhere strictly to the same evaluation protocol across all methods.

Evaluation metrics. For paired benchmarks we report standard full-reference image quality metrics: pixel-level PSNR $\uparrow$ and structural SSIM $\uparrow$ (Wang et al., 2004), the deep perceptual metric LPIPS $\downarrow$ (Zhang et al., 2018), and the distributional similarity metric FID $\downarrow$ (Heusel et al., 2017). For the non-reference benchmarks U45 and C60, we additionally use UIQM$\uparrow$ (Panetta et al., 2015) and UCIQE$\uparrow$ (Yang and Sowmya, 2015).

Table 1: Quantitative comparison on UIEB and LSUI at standardized 256×256 resolution.

Method	UIEB						LSUI					
	PSNR↑	SSIM↑	LPIPS↓	FID↓	UCIQE↑	UIQM↑	PSNR↑	SSIM↑	LPIPS↓	FID↓	UCIQE↑	UIQM↑
UIEWD	14.8806	0.7387	0.3771	92.3536	0.5704	4.0653	17.8399	0.8051	0.3348	53.1968	0.5642	**4.4957**
UWCNN	19.5062	0.8482	0.3080	72.2604	0.5451	3.8867	21.0453	0.8636	0.3264	54.8434	0.5560	4.0106
WaterNet	23.6456	0.9106	<u>0.1623</u>	<u>43.7351</u>	0.5826	**4.2446**	24.9394	0.9015	0.2227	39.2744	<u>0.5857</u>	<u>4.4053</u>
Ushape	24.6389	0.8479	0.1940	48.6329	**0.5857**	<u>4.2037</u>	**29.1284**	<u>0.9172</u>	<u>0.1449</u>	**22.9495**	**0.5865**	4.3093
WF-Diff	<u>24.7457</u>	<u>0.9150</u>	0.1692	53.9730	<u>0.5847</u>	4.1871	28.2709	0.9112	0.1506	33.2456	0.5734	4.3176
Ours	**24.9818**	**0.9316**	**0.1390**	**29.7382**	0.5835	4.0450	<u>28.4551</u>	**0.9246**	**0.1216**	<u>25.9970</u>	0.5743	4.2832

4.2 Results and Discussion

Tables 1–4 summarize the quantitative performance of WaveFlow-UIE under paired restoration, cross-dataset generalization, and inference-time evaluation. In all quantitative tables, bold and underlined values indicate the best and second-best results, respectively. For UIEB and LSUI, the Ours row uses the dataset-matched WaveFlow-UIEB and WaveFlow-LSUI checkpoints. For UFO-120 and EUVP, Ours uses WaveFlow-LSUI and WF-Diff uses WF-Diff-LSUI. We report both distortion-oriented metrics, such as PSNR and SSIM, and perceptual/distributional metrics, such as LPIPS and FID.

Table 1 shows that WaveFlow-UIE is especially strong on perceptual metrics at standardized resolution. On UIEB, our method achieves the best LPIPS and the second-best PSNR, SSIM, and FID, indicating that it produces visually closer reconstructions while remaining competitive in pixel-level fidelity. On LSUI, WaveFlow-UIE obtains the best SSIM and LPIPS and the second-best PSNR and FID. These results suggest that the proposed wavelet-domain flow formulation is effective at preserving structure and perceptual detail across the two main paired benchmarks.

Table 2: Native-resolution quantitative comparison on UIEB and LSUI.

Method	UIEB						LSUI					
	PSNR↑	SSIM↑	LPIPS↓	FID↓	UCIQE↑	UIQM↑	PSNR↑	SSIM↑	LPIPS↓	FID↓	UCIQE↑	UIQM↑
UIEWD	14.3053	0.7507	0.3945	97.1222	0.5622	3.8048	17.4945	0.7853	0.3582	55.0274	0.5600	**4.2795**
UWCNN	19.1571	0.8601	0.3312	74.7177	0.5378	3.3864	20.6892	0.8354	0.3660	54.3170	0.5530	3.7243
WaterNet	23.1406	0.9156	<u>0.1833</u>	<u>36.1890</u>	0.5806	**3.8336**	24.4207	0.8768	0.2558	38.1637	**0.5856**	<u>4.2284</u>
WF-Diff	<u>23.6565</u>	<u>0.9346</u>	0.1963	38.5893	**0.5831**	<u>3.8225</u>	<u>25.9974</u>	**0.8973**	<u>0.1957</u>	<u>27.5106</u>	0.5699	4.0466
Ours	**24.6786**	**0.9350**	**0.1589**	**27.6999**	<u>0.5820</u>	3.6330	**27.2355**	<u>0.8949</u>	**0.1519**	**27.0609**	<u>0.5728</u>	4.1128

Table 2 evaluates the same datasets at their native image resolutions. WaveFlow-UIE achieves the best PSNR, SSIM, LPIPS, and FID on UIEB, and the best PSNR, LPIPS, and FID on LSUI. This indicates that the proposed model is not only tuned to the resized 256×256 setting, but also generalizes well to native-resolution evaluation. The slightly lower UIQM values show that non-reference quality measures do not always align with full-reference and perceptual metrics, especially when color and contrast are enhanced differently by each method.

Table 3 summarizes cross-dataset evaluation on UFO-120, EUVP, U45, and

Table 3: Benchmark results on UFO-120, EUVP, U45, and C60 at 256×256.

Method	UFO-120						EUVP						U45		C60	
	PSNR↑	SSIM↑	LPIPS↓	FID↓	UCIQE↑	UIQM↑	PSNR↑	SSIM↑	LPIPS↓	FID↓	UCIQE↑	UIQM↑	UCIQE↑	UIQM↑	UCIQE↑	UIQM↑
UIEWD	19.4465	0.8122	0.3237	64.2594	0.5925	**4.4257**	19.3659	0.7946	0.3412	46.1214	0.5797	**4.4398**	0.5647	3.9488	0.5227	4.0000
UWCNN	20.7244	0.8625	0.2909	74.2507	0.5905	4.0840	21.5804	0.8521	0.3238	51.8359	0.5778	4.1469	0.5287	3.8195	0.5210	3.5638
WaterNet	24.2772	0.8780	0.2591	52.9071	**0.6010**	4.3494	24.4686	0.8644	0.2876	39.9792	**0.5931**	4.4131	0.5755	**4.2914**	**0.5637**	**4.0146**
Ushape	29.5540	0.9111	0.1792	38.8640	0.5997	4.2437	**31.4794**	**0.9030**	0.1634	**22.9462**	0.5886	4.2795	**0.5860**	4.2262	0.5585	3.8208
WF-Diff	**30.6599**	0.9044	0.1336	36.0713	0.5877	4.2971	29.9486	0.8945	0.1589	23.0850	0.5800	4.3459	–	–	–	–
Ours	30.3101	**0.9146**	**0.1138**	**33.7577**	0.5904	4.2741	29.2063	0.8931	**0.1503**	26.3554	0.5826	4.3222	0.5626	4.0394	0.5501	3.7271

C60 at standardized 256×256 resolution. Using the LSUI-trained checkpoint, WaveFlow-UIE generalizes strongly on the paired benchmarks: on UFO-120 it achieves the best SSIM, LPIPS, and FID while remaining second-best in PSNR, indicating strong preservation of structure and perceptual quality under domain shift. On EUVP, our method obtains the best LPIPS, showing that its outputs remain perceptually close to the reference images, although U-shape and WF-Diff are stronger on PSNR, SSIM, and FID. On the non-reference sets U45 and C60, the mixed-data WaveFlow-BOTH checkpoint remains competitive but does not lead the UIQM/UCIQE rankings, suggesting that WaveFlow generalizes more reliably on paired cross-dataset benchmarks than on fully reference-free challenge images.

Table 4: Inference time comparison on UIEB in milliseconds per image. Lower is better.

Method	UIEB	
	Native Time↓	256 Time↓
UIEWD	24.28	3.32
UWCNN	141.12	12.50
WaterNet	118.50	4.68
Ushape	–	28.01
WF-Diff	3392.32	301.30
Ours	444.83	79.31

Table 4 compares inference time on UIEB. WaveFlow-UIE is substantially faster than WF-Diff, reducing the native-resolution runtime from 3392.32 ms to 444.83 ms and the 256 × 256 runtime from 301.30 ms to 79.31 ms. This supports the motivation for replacing the two-stage WF-Diff pipeline with a single flow-based pipeline. However, WaveFlow-UIE remains slower than lightweight feed-forward methods such as UIEWD and WaterNet, mainly because it still performs multiple flow update steps during inference.

4.2.1 Ablation Study

Table 5 reports the ablation study on UIEB at standardized 256×256 resolution. The full WaveFlow-UIE model achieves the best overall balance across PSNR,

Table 5: UIEB ablation results at standardized 256×256 resolution.

Configuration	PSNR↑	SSIM↑	LPIPS↓	FID↓	UCIQE↑	UIQM↑	Time↓ (ms)
Full model (K=5)	24.9818	0.9316	0.1390	29.7382	0.5835	4.0450	79.31
K=1 single-step	24.9382	0.9306	0.1403	30.6582	0.5827	4.0638	19.55
No physics prior	24.5000	0.9234	0.1469	35.7000	0.5853	4.0912	–
No frequency weighting	24.0902	0.9298	0.1426	33.8364	0.5852	4.0270	–
No LPIPS loss	24.0288	0.9047	0.1574	35.1146	0.5753	4.2561	–
No Lab color loss	24.1482	0.9220	0.1848	44.0632	0.5819	4.0640	–

SSIM, LPIPS, and FID, showing that the final design is not driven by any single component alone. Removing the physics prior reduces performance across all main restoration metrics, indicating that explicit transmission and ambient-light conditioning helps the model correct underwater degradation more consistently. Removing the frequency weighting also hurts performance, especially in PSNR and FID, which suggests that emphasizing high-frequency wavelet errors is important for preserving detail while maintaining global fidelity. The no-LPIPS and no-Lab variants cause the largest perceptual degradation, with the no-Lab model producing the worst LPIPS and FID among the trained loss ablations; this shows that both perceptual supervision and Lab-space color consistency are important, and that the color loss is especially critical for stable underwater restoration. Finally, the single-step variant ($K = 1$) remains very close to the full model in quality while reducing inference time from 79.31 ms to 19.55 ms, showing that most of the restoration performance can be retained with a much cheaper inference process.

5 Conclusion

In this work, we presented WaveFlow-UIE, a single-pipeline wavelet-domain flow model for underwater image restoration. The method keeps the main frequency-domain insight of WF-Diff, where low-frequency components mainly capture color and illumination while high-frequency components capture edges and texture details, but removes the need for a separate coarse enhancement network followed by diffusion-based residual refinement. Instead, WaveFlow-UIE directly learns a conditional velocity field that moves degraded wavelet representations toward clean wavelet representations.

Experiments across UIEB, LSUI, UFO-120, and EUVP show that WaveFlow-UIE achieves competitive restoration quality, with strong performance on perceptual metrics such as LPIPS and FID and consistent improvements over WF-Diff in several settings. The runtime comparison also shows that the proposed single-pipeline design is substantially faster than WF-Diff. However, WaveFlow-UIE is still slower than lightweight feed-forward enhancement methods and is not consistently best on all non-reference quality metrics. Future work can focus on reducing the number of flow inference steps, improving efficiency, and strengthening generalization across different underwater domains.

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